Rule-Based Modeling of Cell Signaling

Module 2: Molecular structure and combinatorial complexity

Jim Faeder

MSCBIO 2025 · Introduction to Bioinformatics Programming in Python

Fall 2026

Resources

- **BioNetGen** software, documentation, and tutorials: bionetgen.org
- **VS Code extension**: search the Extensions panel for *BioNetGen*
- **BNG Playground**: write and simulate models in your browser, no installation ruleworld.github.io/bngplayground
- **bngsim**: fast ODE and stochastic simulation from Python bngsim.readthedocs.io · `pip install bngsim bionetgen`
- **Course repository**: models, notebooks, slides and handouts github.com/jrfaeder/RBM_Intro_BioNetGen



Course repository



BNG Playground

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3. read and write **patterns**, and use them to define observables
4. write **reaction rules** and predict which reactions a rule generates
5. describe how BioNetGen **generates a network** from rules, and when network-free simulation is needed instead

Part 1

From species to molecules

Where Module 1 left off

Our cytokine–STAT–SOCS model had **12 species**, each named by hand:

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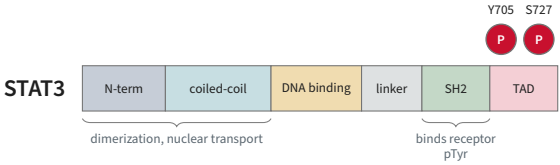
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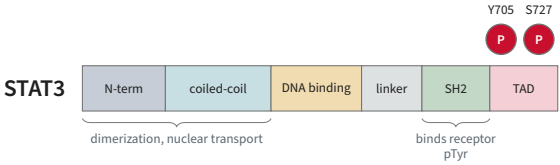
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Each of these is false for real cytokine receptors. Relaxing any of them requires new species, and new copies of every reaction they take part in.

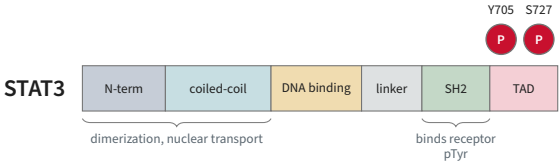
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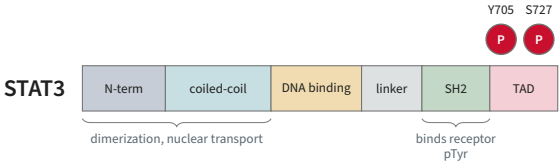
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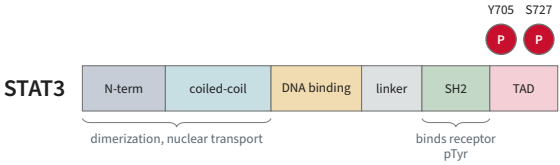
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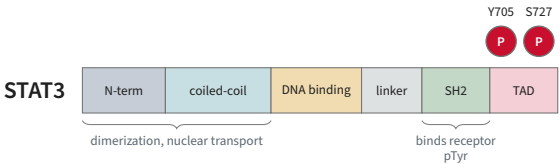
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n independent two-state sites

$\Rightarrow$ 2^n possible states of one molecule

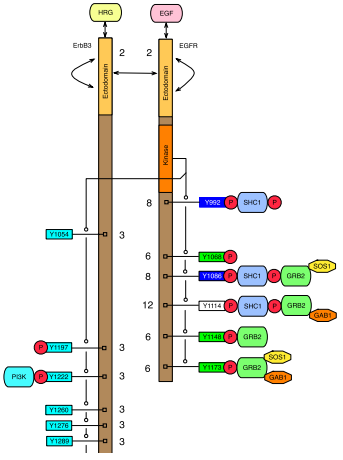
and complexes multiply the counts of their parts

- Each protein has several **binding domains** and **modification sites**.
- Sites can be occupied or modified *in combination*.
- Most interactions depend on only one or two sites: SOCS binding does not care whether S727 is phosphorylated.
- Naming every combination is **combinatorial complexity**: a few interactions imply a huge number of species.

Combinatorial complexity in a realistic model of EGFR signaling

A model of the ErbB receptor family with its known binding partners:

- ErbB3:ErbB1 dimer: $> 3.8 \times 10^9$ states

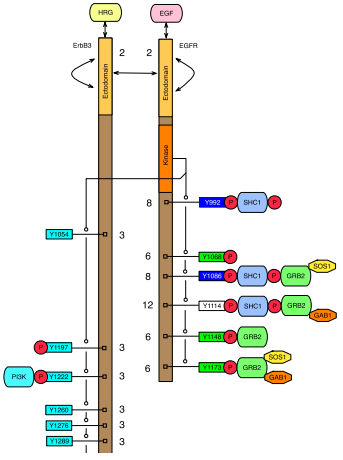


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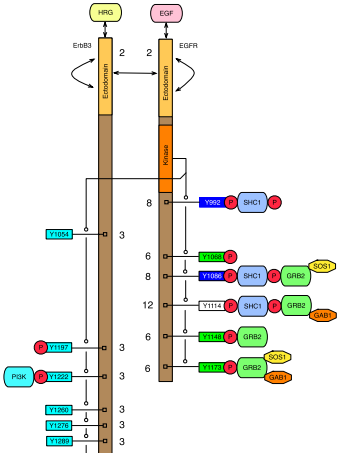
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Yet the model is built from a manageable number of **rules**, in the tens to hundreds.

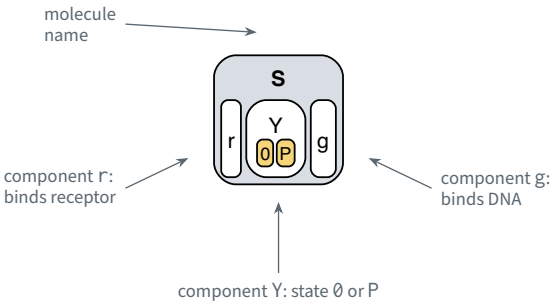


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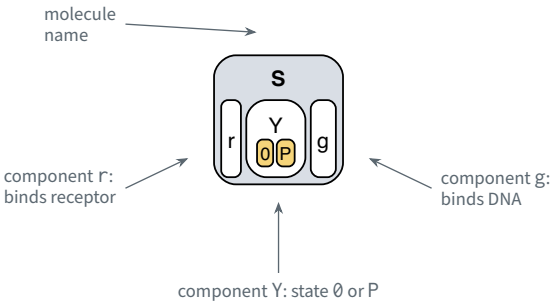
Part 2

Structured molecules in BNGL

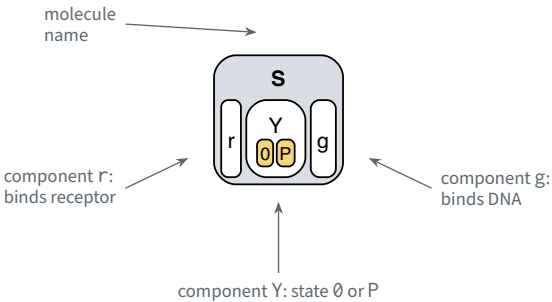
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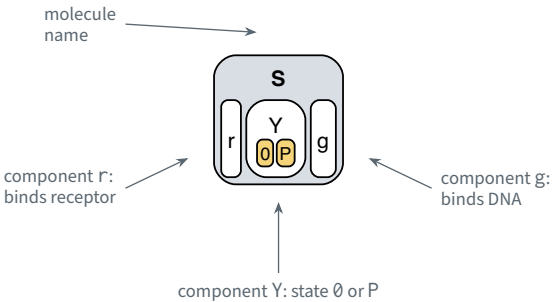
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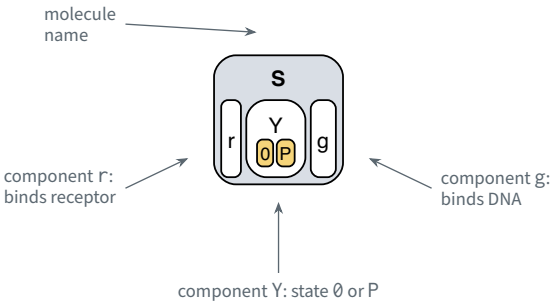
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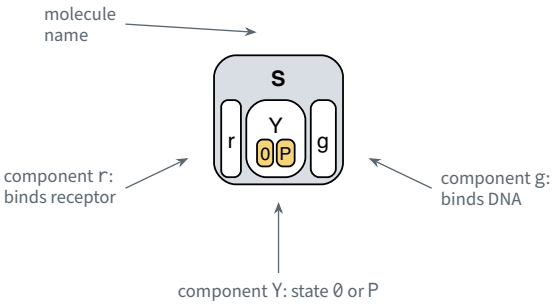
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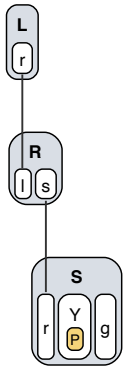
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- In the diagrams: molecules gray, components white, states yellow.

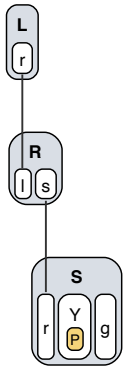
Complexes are molecules joined by bonds



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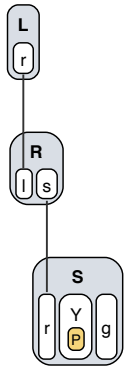


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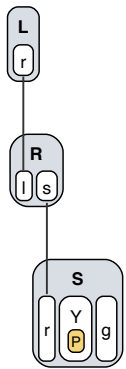


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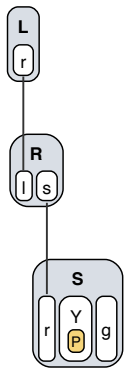


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- A species is now a **graph**: molecules contain components, and bonds join components. BioNetGen recognizes when two strings describe the same graph.

Patterns select sets of species

A **pattern** lists only the parts of a complex you care about. **Don’t write, don’t care**: anything not mentioned can be in any state.

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`R(l!+,s)`	✓	×	×	×	×	×	… with docking site free

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- Matching is done on the graph, so the pattern can match a molecule anywhere inside a complex.

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`Molecules` VS. `Species`

- `Molecules` counts every *match*. A STAT dimer with two pTyr contributes 2 to `S(Y~P)`.
- `Species` counts each matching *complex* once.

```
Species pSTATdimer S(d!1).S(d!1)
```

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- **Reactant patterns** (left) say what must be present: a ligand-bound receptor with a free docking site, and an unphosphorylated STAT with a free `r` site.
- **Product patterns** (right) say what changes: a bond forms between `s` and `r`.
- Everything not mentioned is carried along unchanged.
- Rates follow mass action, with the *rule's* rate constant (times a statistical factor for identical molecules or sites).

Rules transform patterns

```
Sbind:  R(l!+,s) + S(r,Y~0) <-> R(l!+,s!1).S(r!1,Y~0)  kSon, kSoff
```

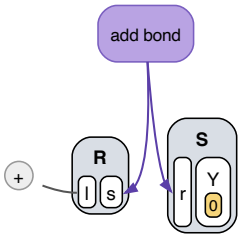
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What a rule can do

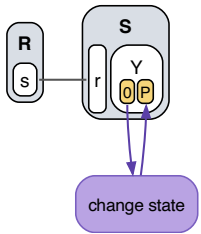
form a bond	`A(b) + B(a) -> A(b!1).B(a!1)`
break a bond	`A(b!1).B(a!1) -> A(b) + B(a)`
change a state	`S(Y~0) -> S(Y~P)`
make a molecule	`G(s!+) -> G(s!+) + mSOCS()`
delete one	`SOCS(r) -> 0`

Rule diagrams: what a rule changes, and where it applies

```
Sbind:  R(l!+,s) + S(r,Y~0) <-> R(l!+,s!1).S(r!1,Y~0)  kSon, kSoff
Sact:   R(s!1).S(r!1,Y~0) -> R(s!1).S(r!1,Y~P)         kact
```



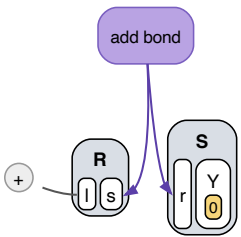
Sbind



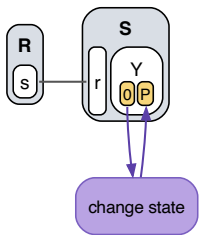
Sact

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```



Sbind

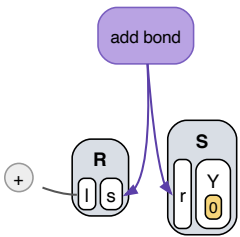


Sact

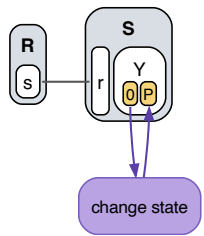
- **Purple** nodes are **operations**, with arrows to what they change: a new bond, or `Y` from `0` to `P`.

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Sact:   R(s!1).S(r!1,Y~0) -> R(s!1).S(r!1,Y~P)         kact
```



Sbind

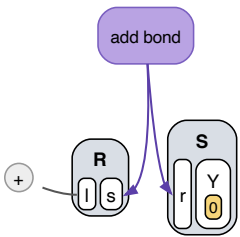


Sact

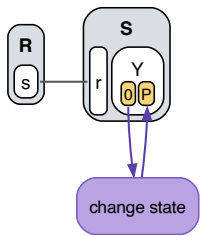
- **Purple** nodes are **operations**, with arrows to what they change: a new bond, or `Y` from `0` to `P`.
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```



Sbind



Sact

- **Purple** nodes are **operations**, with arrows to what they change: a new bond, or `Y` from `0` to `P`.
- Everything else is **context**: gray lines are bonds that must exist, `+` is a bond to any partner (`l!+`).
- A component drawn with no line must be free. One the rule does not mention (`g`) is not drawn.

Generated with `visualize({type=>"ruleviz_operation"})` (Sekar et al., 2017).

Part 3

The Module 1 model as rules

Molecule types, seed species, and observables

```
begin molecule types
  L(r)
  R(l,s)
  S(r,Y~0~P,g)
  G(s)
  mSOCS()
  SOCS(r)
end molecule types
begin seed species
  L(r)        L0
  R(l,s)      R0
  S(r,Y~0,g)  S0
  G(s)        G0
end seed species
begin observables
  Molecules pSTAT  S(Y~P)
  Molecules SOCS   SOCS()
end observables
```

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```
begin molecule types
  L(r)
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begin seed species
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  R(l,s)      R0
  S(r,Y~0,g)  S0
  G(s)        G0
end seed species
begin observables
  Molecules pSTAT  S(Y~P)
  Molecules SOCS   SOCS()
end observables
```

- Six molecule types replace twelve species names.

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end molecule types
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  R(l,s)      R0
  S(r,Y~0,g)  S0
  G(s)        G0
end seed species
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  Molecules SOCS   SOCS()
end observables
```

- Six molecule types replace twelve species names.
- **Seed species** are the complexes present at $t = 0$. Every other species is generated from them by the rules.

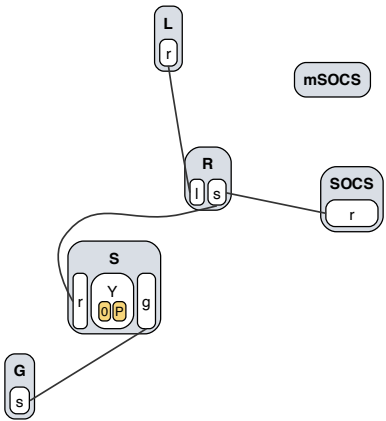
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  L(r)
  R(l,s)
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  G(s)
  mSOCS()
  SOCS(r)
end molecule types
begin seed species
  L(r)        L0
  R(l,s)      R0
  S(r,Y~0,g)  S0
  G(s)        G0
end seed species
begin observables
  Molecules pSTAT  S(Y~P)
  Molecules SOCS   SOCS()
end observables
```

- Six molecule types replace twelve species names.
- **Seed species** are the complexes present at $t = 0$. Every other species is generated from them by the rules.
- The receptor has one docking site `s` that STAT and SOCS compete for. This is the Module 1 assumption.

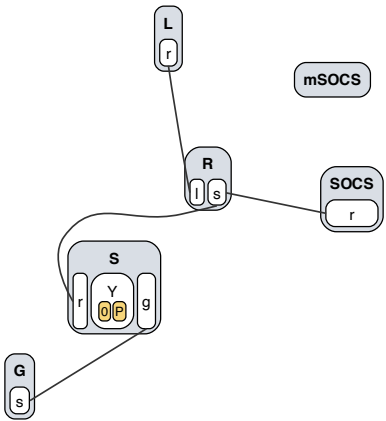
File: `models/stat_rbm1_rules.bngl`

The contact map: what can bind what



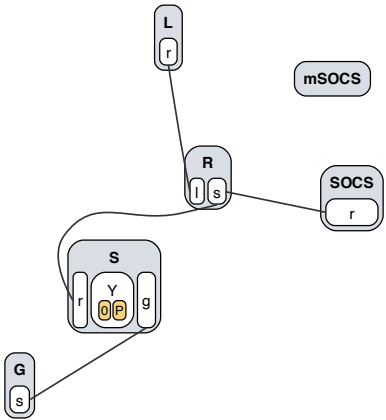
- Each molecule type appears once, with its components and states.

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- Each line joins two components that some rule can bind.

The contact map: what can bind what



- Each molecule type appears once, with its components and states.
- Each line joins two components that some rule can bind.
- It shows what is *possible*, not *when*: STAT and SOCS both bind `s`, but the map does not say that only ligand-bound receptors recruit. That is the rules' context.

Generated with `visualize({type=>"contactmap"})`.

The rules

```
LRbind:     L(r) + R(l,s) <-> L(r!1).R(l!1,s)              kon, koff
Sbind:      R(l!+,s) + S(r,Y~0) <-> R(l!+,s!1).S(r!1,Y~0)  kSon, kSoff
Sact:       R(s!1).S(r!1,Y~0) -> R(s!1).S(r!1,Y~P)         kact
Srelease:   R(s!1).S(r!1,Y~P) -> R(s) + S(r,Y~P)           kdiss
Sdephos:    S(r,g,Y~P) -> S(r,g,Y~0)                       kdephos
Gbind:      S(r,Y~P,g) + G(s) <-> S(r,Y~P,g!1).G(s!1)      kGon, kGoff
Transcribe: G(s!+) -> G(s!+) + mSOCS()                     ktx
mRNAdeg:    mSOCS() -> 0                                   kmdeg
Translate:  mSOCS() -> mSOCS() + SOCS(r)                   ktl
SOCSdeg:    SOCS(r) -> 0                                   kSOCSdeg
SOCSbind:   R(l!+,s) + SOCS(r) <-> R(l!+,s!1).SOCS(r!1)    kSOCSon, kSOCSoff
```

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LRbind:     L(r) + R(l,s) <-> L(r!1).R(l!1,s)              kon, koff
Sbind:      R(l!+,s) + S(r,Y~0) <-> R(l!+,s!1).S(r!1,Y~0)  kSon, kSoff
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mRNAdeg:    mSOCS() -> 0                                   kmdeg
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SOCSbind:   R(l!+,s) + SOCS(r) <-> R(l!+,s!1).SOCS(r!1)    kSOCSon, kSOCSoff
```

Context encodes the Module 1 assumptions. `R(l,s)` in `LRbind`: ligand binds and leaves only when the docking site is empty. `R(l!+,s)` in `Sbind` and `SOCSbind`: only ligand-bound receptors recruit.

Network generation: rules applied to seed species

1. Start with the seed species.

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4. Repeat until nothing new appears.

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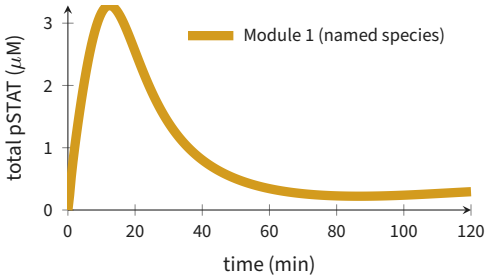
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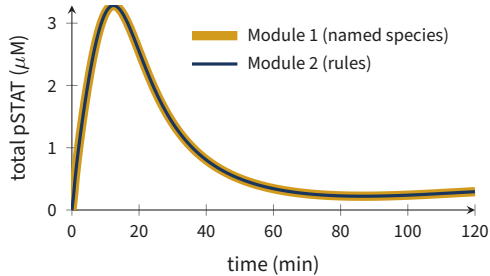


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Result: the same 12 species and 15 reactions as Module 1.



The curves are identical.

Part 4

Relaxing the assumptions

1. Ligand can leave any receptor complex

Remove the context `s` from the ligand rule:

```
LRbind:  L(r) + R(l) <-> L(r!1).R(l!1)  kon, koff
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```

Docking and activation still need bound ligand, but undocking does not, so each reversible rule is split in two:

```
Sbind:   R(l!+,s) + S(r,Y~0) -> R(l!+,s!1).S(r!1,Y~0)  kSon
Sunbind: R(s!1).S(r!1,Y~0) -> R(s) + S(r,Y~0)          kSoff
Sact:    R(l!+,s!1).S(r!1,Y~0) -> R(l!+,s!1).S(r!1,Y~P)  kact
```

(and likewise for SOCS)

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Sact:    R(l!+,s!1).S(r!1,Y~0) -> R(l!+,s!1).S(r!1,Y~P)  kact
```

(and likewise for SOCS)

One rule, many reactions

`LRbind` now generates 4 binding reactions: ligand binds R alone, R·S, R·S_p, or R·SOCS.

	species	reactions
before	12	15

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One rule, many reactions

`LRbind` now generates 4 binding reactions: ligand binds R alone, R·S, R·S_p, or R·SOCS.

	species	reactions
before	12	15
after	**15**	**24**

2. SOCS binds its own site and blocks kinase activity

Give the receptor a separate SOCS site. SOCS no longer blocks docking; it blocks *activation*:

```
R(l,s,socs)

Sact:       R(l!+,socs,s!1).S(r!1,Y~0) ->
              R(l!+,socs,s!1).S(r!1,Y~P)  kact
SOCSbind:   R(l!+,socs) + SOCS(r) ->
              R(l!+,socs!1).SOCS(r!1)     kSOCSon
SOCSunbind: R(socs!1).SOCS(r!1) -> R(socs) + SOCS(r)  kSOCSoff
```

`socs` free in `Sact` means: activation only when SOCS is absent.

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Closer to how SOCS proteins work: SOCS1 inhibits JAK directly, and SOCS3 inhibits JAK while bound to the receptor.

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Now a receptor can carry STAT *and* SOCS at the same time.

	species	reactions
before	15	24

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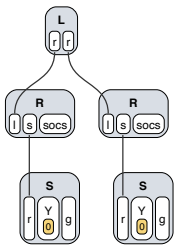
	species	reactions
before	15	24
after	**19**	**39**

3. Cytokines bring two receptor chains together

In the simplest case, a bivalent ligand `L(r,r)` crosslinks two identical receptor chains, whose JAKs activate each other, so STAT is activated only in a dimer:

```
L(r,r)

LRbind:    L(r,r) + R(l) <-> L(r!1,r).R(l!1)      kon, koff
Crosslink: L(r!+,r) + R(l) <-> L(r!+,r!1).R(l!1)  kx, koff
Sact:      L(r!+,r!2).R(l!2,socs,s!1).S(r!1,Y~0) ->
             L(r!+,r!2).R(l!2,socs,s!1).S(r!1,Y~P)  kact
```

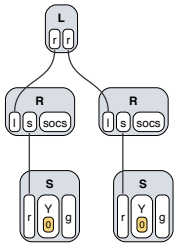


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Sact:      L(r!+,r!2).R(l!2,socs,s!1).S(r!1,Y~0) ->
             L(r!+,r!2).R(l!2,socs,s!1).S(r!1,Y~P)  kact
```



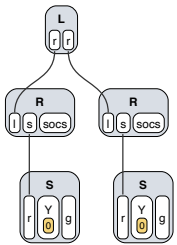
- BioNetGen accounts for the two identical ligand sites automatically (statistical factor of 2 in `LRbind`).

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```



- BioNetGen accounts for the two identical ligand sites automatically (statistical factor of 2 in `LRbind`).
- Excess bivalent ligand occupies chains one at a time and blocks crosslinking, so we lower L_0 from 10 to 1 μM.

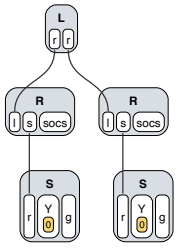
3. Cytokines bring two receptor chains together

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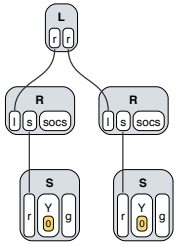
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	species	reactions
before	19	39
after	**40**	**188**

`Crosslink` alone generates 36 reactions in each direction.

4. STATs act as dimers

Add a dimerization site. Free pSTAT dimerizes, only dimers bind the SOCS promoter, and only monomers are dephosphorylated:

```
S(r,Y~0~P,d,g)

Sdimer:  S(r,Y~P,d) + S(r,Y~P,d) <->
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Gbind:   S(d!+,g) + G(s) <-> S(d!+,g!1).G(s!1)  kGon, kGoff
Sdephos: S(r,d,g,Y~P) -> S(r,d,g,Y~0)          kdephos
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5. A second phosphorylation site doubles everything

STAT3 is also phosphorylated on Ser727. We assume this is independent of Tyr705; context could change that. One new component and *one* rule:

```
S(r,Y~0~P,S727~0~P,d,g)

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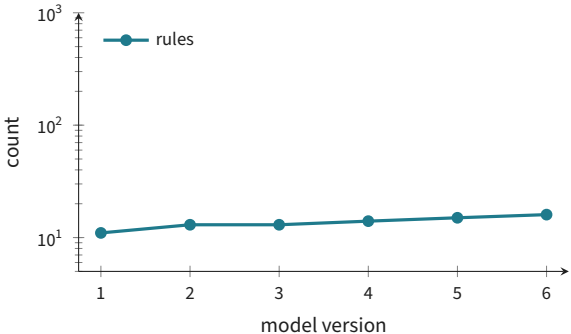
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	species	reactions
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after	**95**	**663**

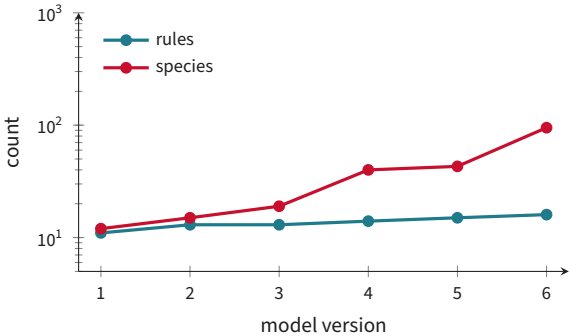
`Sser` alone generates 59 forward and 59 reverse reactions.

Rules grow linearly; the network grows combinatorially



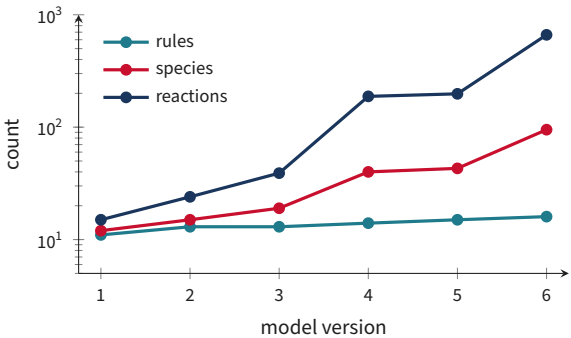
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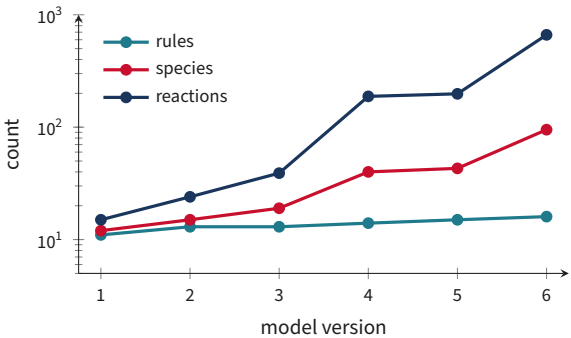
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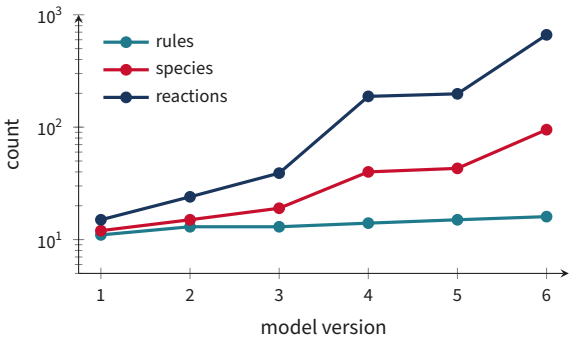
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From 11 to 16 rules, the network grows from 15 to 663 reactions.

Rules grow linearly; the network grows combinatorially

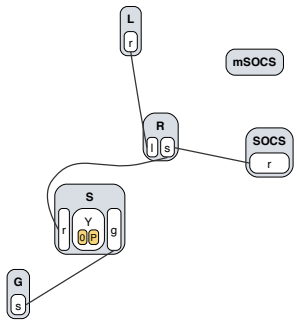


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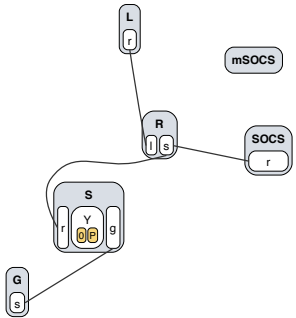
Writing these 663 reactions, and the ODEs for 95 species, by hand would be slow and error-prone. **Rules let us write down the mechanism and let the software do the bookkeeping.**

The contact map grows like the rules

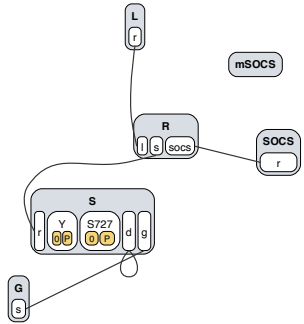


version 1: 8 components, 4 possible bonds

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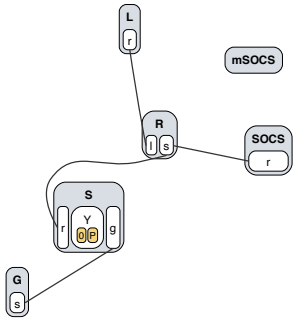


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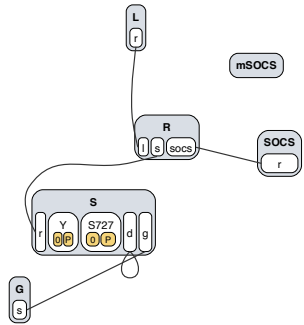


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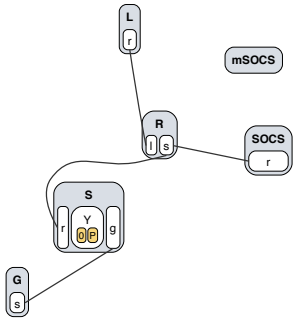
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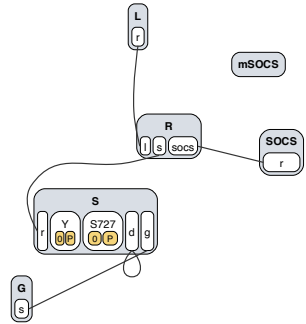
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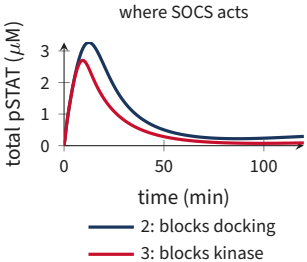
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The contact map, like the rules, stays small enough to read at a glance.

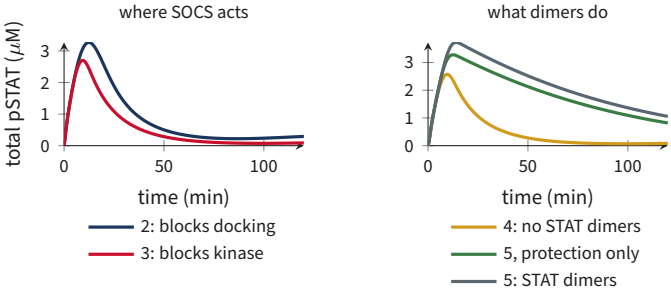
Mechanistic details change the predictions

Compare models that differ in *one* mechanism.



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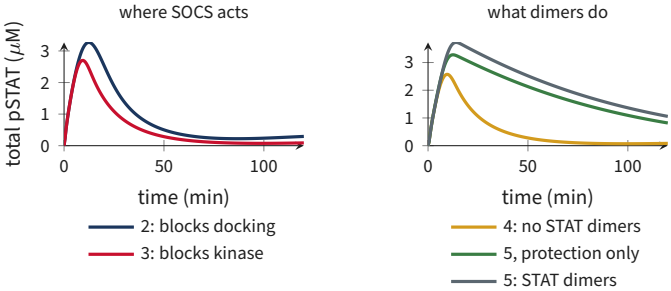
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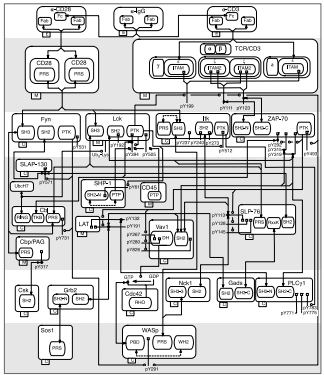
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- Experiments decide which mechanism is right. Rules make the alternatives easy to build and compare.

Part 5

Scaling up

Rules capture knowledge at large scale

T cell receptor model



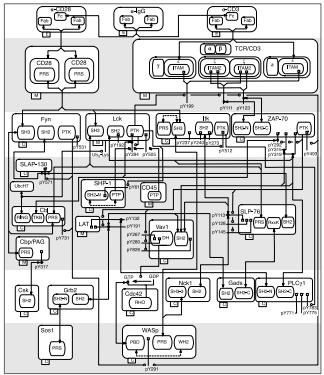
20 molecule types
129 rules

Number of species is too large for network generation!

A rule-based model of T cell receptor signaling: 20 molecule types and 129 rules.

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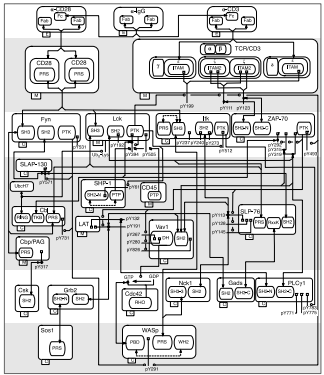
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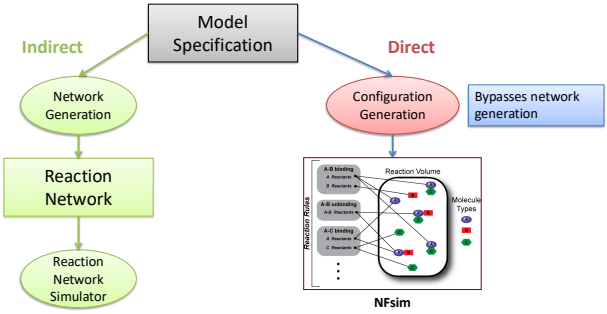
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Rule-based models also serve as precise, extensible, visualizable representations of what is known about a pathway.

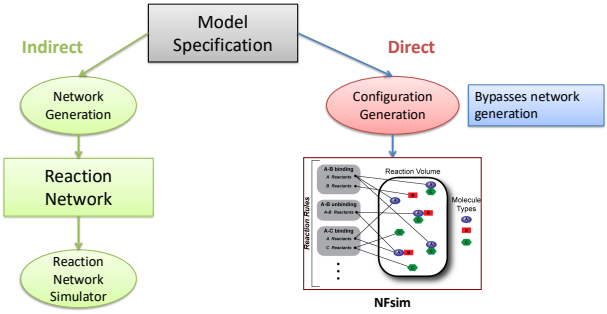
Chylek et al., *PLOS One* (2014)

Two ways to simulate a rule-based model



Indirect: generate the network, then integrate ODEs or run SSA on it. Fast when the network is small. This is what we have done so far.

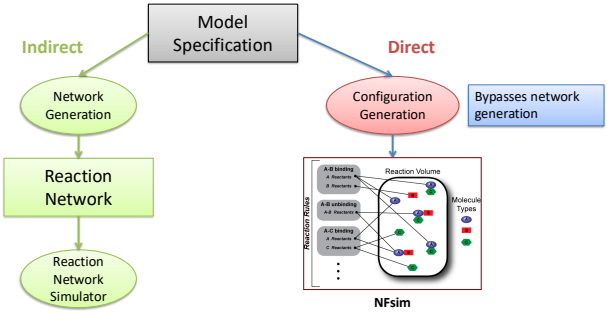
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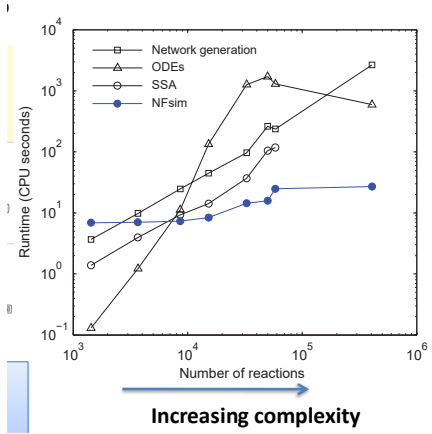


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`simulate({method=>"nf",...})` runs NFsim.

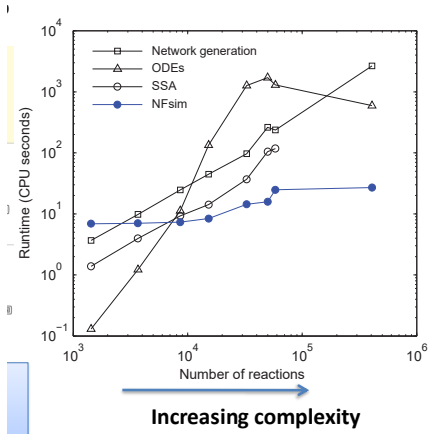
Network-free simulation scales with the rules, not the network



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Sneddon, Faeder, and Emonet, *Nat. Methods* **8**, 177 (2011)

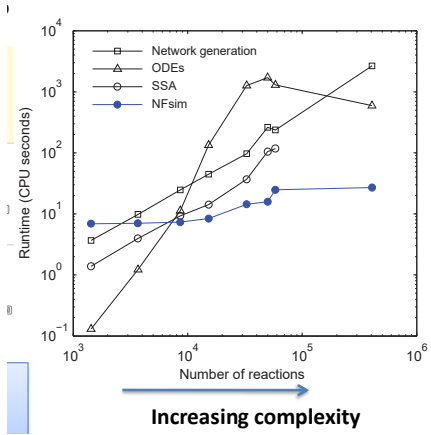
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- The cost of network generation and ODE integration grows with the number of reactions.
- NFsim's cost per event depends on the number of rules and molecules, so it handles models whose networks are huge or infinite.
- Trade-off: NFsim is stochastic and works with molecule counts, so results are noisy and concentrations must be converted to numbers.

Sneddon, Faeder, and Emonet, *Nat. Methods* **8**, 177 (2011)

Try it yourself

Using `models/stat_rbm*.bngl`:

1. In `stat_rbm1_rules.bngl`, change `R(l,s)` to `R(l)` in `LRbind`. Generate the network. Which new species appear, and why?
2. Write an observable for receptors that carry both STAT and SOCS in `stat_rbm3_socs_site.bngl`.
3. In `stat_rbm4_receptor_dimer.bngl`, vary `L0` from 0.01 to 100. Plot peak pSTAT against `L0`. Explain the shape.
4. Remove the `r` context from `Sdimer` in `stat_rbm5_stat_dimer.bngl` so receptor-bound pSTAT can dimerize. What happens when you generate the network? Try `generate_network({max_agg=>6})`.

No local setup yet? Paste any of these models into BNG Playground (`ruleworld.github.io/bngplayground`) and run it in your browser.

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- BioNetGen **generates the network** from rules for ODE/SSA simulation; **NFsim** simulates models whose networks are too large to generate.